

Assignment 1 – Command Line Arguments

Description:

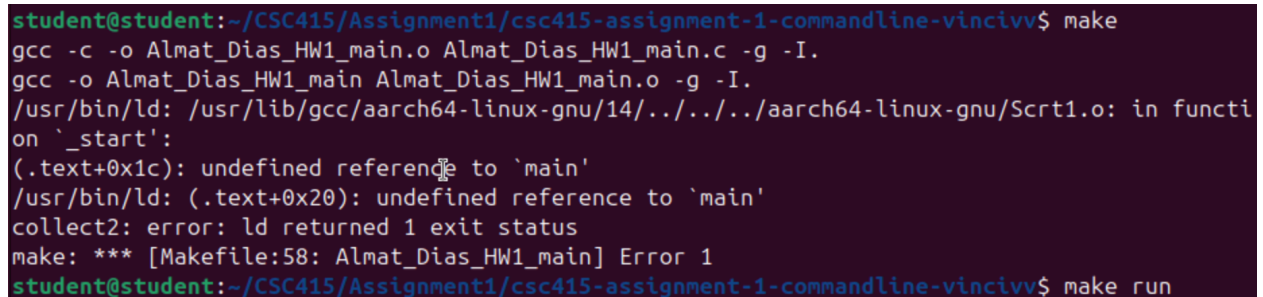
This assignment is to write a C program that accepts arguments via the command line and then displays each of those arguments to the terminal along with how many arguments there are.

Approach:

My approach is first to create a C file and initialize the main function with argc and argv as arguments, which are the number of arguments and the content of the arguments as an array, taken from the command line. Then print them in the terminal. Also, I need to change the Makefile, so that it compiles right.

Issues and Resolutions:

My first issue was that the program would not run. I was trying to do make, but it gave me an error. Here is the screenshot:



```
student@student:~/CSC415/Assignment1/csc415-assignment-1-commandline-vincivv$ make
gcc -c -o Almat_Dias_HW1_main.o Almat_Dias_HW1_main.c -g -I.
gcc -o Almat_Dias_HW1_main Almat_Dias_HW1_main.o -g -I.
/usr/bin/ld: /usr/lib/gcc/aarch64-linux-gnu/14/../../../../aarch64-linux-gnu/Scrt1.o: in function '_start':
(.text+0x1c): undefined reference to `main'
/usr/bin/ld: (.text+0x20): undefined reference to `main'
collect2: error: ld returned 1 exit status
make: *** [Makefile:58: Almat_Dias_HW1_main] Error 1
student@student:~/CSC415/Assignment1/csc415-assignment-1-commandline-vincivv$ make run
```

I resolved it by adding the main function to my c file. The problem was that it did not see the main function, so it didn't know what to compile.

My next issue was that I did not know how to get the arguments from the command line. First, my idea was to import the makefile to my main.c file, and get the arguments from there, from the RUNOPTIONS field. However, it didn't work. Then I tried to find a solution online. I found in stackoverflow that you can pass the number and the array of arguments into the main function, so that is what I did

Analysis: (If required for the assignment)

Screen shot of compilation:

```
student@student:~/CSC415/Assignment1/csc415-assignment-1-commandline-vincivv$ make  
gcc -c -o Almat_Dias_HW1_main.o Almat_Dias_HW1_main.c -g -I.  
gcc -o Almat_Dias_HW1_main Almat_Dias_HW1_main.o -g -I.  
student@student:~/CSC415/Assignment1/csc415-assignment-1-commandline-vincivv$
```

Screen shot(s) of the execution of the program:

```
student@student:~/CSC415/Assignment1/csc415-assignment-1-commandline-vincivv$ make run  
./Almat_Dias_HW1_main Hello World
```

```
The argument count is: 3  
Argument 0 is: ./Almat_Dias_HW1_main  
Argument 1 is: Hello  
Argument 2 is: World
```

```
student@student:~/CSC415/Assignment1/csc415-assignment-1-commandline-vincivv$
```

```
student@student:~/CSC415/Assignment1/csc415-assignment-1-commandline-vincivv$ make run RUNOPTIONS="Hello CSC415 here are more run options"  
./Almat_Dias_HW1_main Hello CSC415 here are more run options
```

```
The argument count is: 8  
Argument 0 is: ./Almat_Dias_HW1_main  
Argument 1 is: Hello  
Argument 2 is: CSC415  
Argument 3 is: here  
Argument 4 is: are  
Argument 5 is: more  
Argument 6 is: run  
Argument 7 is: options
```

```
student@student:~/CSC415/Assignment1/csc415-assignment-1-commandline-vincivv$
```